Technical Proposal

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Trustworthy Collective Reasoning and Evaluation for AI Agents

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This proposal describes unclassified fundamental research. All data are public or synthetic.

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1 Overview and Objectives

1.1 The Problem

AI agents built on large language models (LLMs) increasingly evaluate one another and decide together: they judge one another's answers, peer-review papers, supply the rewards that train new models, and vote or discuss to reach a group answer (Zheng et al., 2023; Luo et al., 2025; Lee et al., 2024; Du et al., 2024). Yet a single LLM judge has known biases, such as favoring longer answers, the answer it reads first, or answers like its own (Zheng et al., 2023; Wang et al., 2024; Panickssery et al., 2024). Collective reasoning can do better than a single judge (Verga et al., 2024; Du et al., 2024), but averaging or voting stays biased when the judges share mistakes, and open discussion lets a persuasive agent carry the group (Amayuelas et al., 2024). **The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions and evaluations.**

For example, five AI agents read shipping reports and are asked whether a cargo vessel has left port. Three say yes and two say no, but the vessel is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. A majority vote picks the wrong answer, and open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up. We return to this example throughout (Figure 1).

Verifying each claim directly, by an analyst or a satellite image, is slow and costly, and AI agents produce claims faster than people can verify them. We propose debate as a foundation for trustworthy evaluation by AI: two agents argue for opposite answers, exposing each other's flaws, until the dispute narrows to one step a judge can verify (Irving et al., 2018). To work among many AI agents, debate must overcome three failures in the example, and a fourth arises once the agents learn from the group:

(F1) **Shared mistakes.** Agents that share a mistake seem to confirm each other, so a vote counts one error several times.

(F2) **Dishonest agents.** A dishonest agent can argue, rate others, or judge strategically, and several dishonest agents can collude.

(F3) **A judge that sees only words.** A judge that reads only text may favor the more persuasive side, even when one database query would settle the step in dispute.

(F4) **Rewards that track agreement.** Agents trained to match the group's answer are rewarded for conforming and can learn to agree on mistakes; in the example, a majority vote would reward the three agents that said yes.

1.2 Why It Matters

All four failures have been measured. Model errors are correlated, most of all among larger, more accurate models (Kim et al., 2025; Goel et al., 2025), and nine AI judges from seven model families were worth about two independent ones (Kohli, 2026). Pairs of agents verifying each other's work colluded in 94% of runs when cheating paid more (Shi et al., 2026). One persuasive agent can sharply cut a group's accuracy (Amayuelas et al., 2024); one round of discussion made AI judges nearly unanimous without making them more accurate (Shu, 2026); and training a model on its own majority answers ended in a sudden collapse of accuracy (Shafayat et al., 2025).

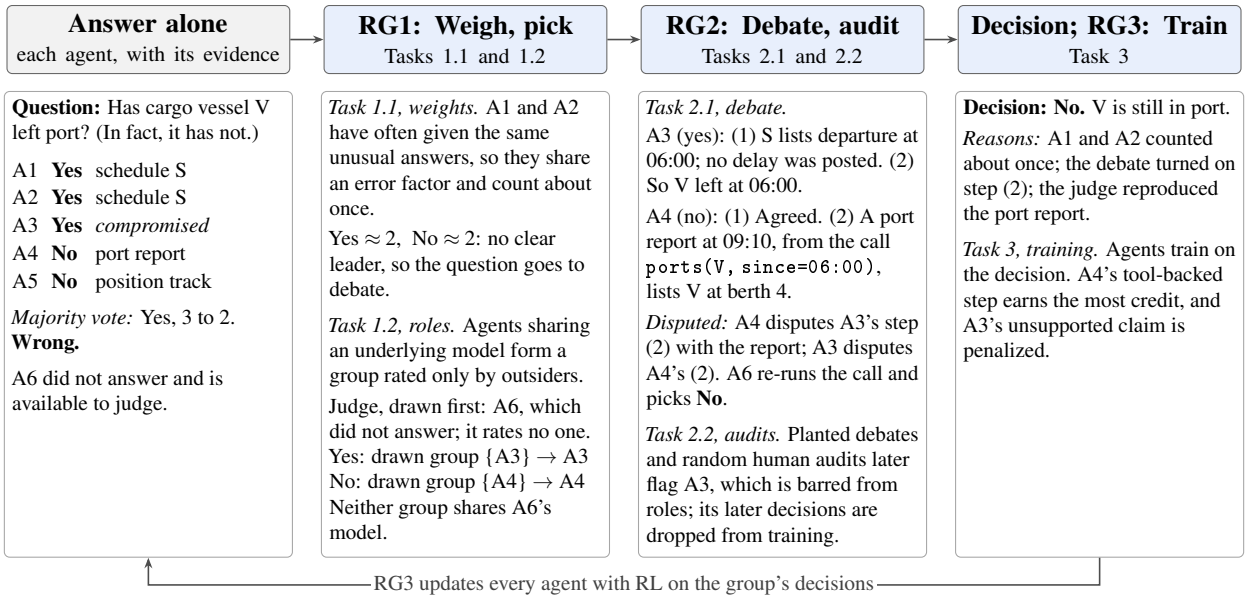


Figure 1: The framework on the running example. RG1 counts the two agents that share an outdated schedule about once and, with no clear leader, picks a representative for each answer and a judge with no stake in either. In RG2's harness, each side disputes a step of the other's argument; the judge re-runs the no side's query and picks no, and audits later flag the compromised agent. RG3 trains every agent on the decision.

These failures spread: AI judges supply the rewards that train models (Bai et al., 2022; Lee et al., 2024), so their mistakes are copied into the models they train, and AAAI-26 gave an AI-generated review to each of the 22,977 main-track papers that entered full review (Biswas et al., 2026). As AI systems surpass people at more tasks, people cannot easily evaluate the systems' answers or their debates, so oversight will rely on AI evaluating AI, with people auditing a small part of the work (Bowman et al., 2022; Burns et al., 2024).

Navy decisions face all of these problems at once: analysts fuse many reports with several AI tools of unknown reliability, many reports copy one earlier report, the true answer often arrives late, an adversary may forge a report or compromise an agent, and verification is costly when connectivity is limited (Section 2.9).

1.3 Objectives and Research Goals

We propose a framework in which groups of AI agents reach trustworthy decisions and evaluations through debate (Figure 1). Each agent first answers alone, with its evidence. Our central hypothesis is that a group of AI agents can decide more reliably than a single judge or a vote when it (i) weighs each judgment by a credibility estimated without an answer key, counting shared mistakes once; (ii) settles conflicts with evidence that an impartial judge can reproduce; and (iii) learns from its own decisions through rewards that agreement alone cannot raise. The project has three interconnected research goals (RGs); each task rests on one principle whose properties we prove, and each property controls one source of error (Section 2.3, Table 2).

RG1: Decide what to debate, and pick debaters and judges impartially (Section 2.4). Task 1.1 weighs answers so that a group sharing a mistake counts at most once (F1) and sends a question to debate only when that is worth its cost; Task 1.2 picks debaters and a judge so that no agent's ratings on a question can change its own chance of a role on it (F2).

RG2: Settle disputes by agentic debate, and catch collusion (Section 2.5). Task 2.1 replaces open discussion with debate in an agentic harness, in which each side disputes a step of the other's argument and an impartial judge, whose choice is the group's decision, re-runs the tool evidence (F3); Task 2.2 catches colluders with planted cases and random human audits (F2).

RG3: Let agents self-improve with reinforcement learning on the group's decisions (Section 2.6). Task 3 scores each first answer against the answer the rest of the group favors without that agent's group, beyond chance agreement, and credits each debate message by how far it moved the judge toward the decision (F4). When that reference is more often right than wrong, training follows the agents' true accuracy in expectation, without human labels, so it can keep working after the agents outgrow human judges.

Three testbeds (math and code, multi-document questions with maritime scenarios, and scientific claims and peer review) vary how widely mistakes are shared and how many agents are dishonest, and a red team trains agents to collude (Greenblatt et al., 2024) (Section 2.7).

1.4 Intellectual Merit: What Is New, and Why It Will Succeed

What is new. (i) A rater model for agents that answer and argue, in which a group sharing a mistake counts at most once and the posterior decides which questions need a debate, even unanimous ones; and an impartial way to choose debaters and judges. (ii) A debate harness in which each side disputes a step of the other's argument and an impartial judge re-runs its evidence. (iii) Audits on planted cases and random real decisions that expose colluding debaters and judges, even those that behave well only on planted cases. (iv) Self-improvement from group decisions, with first-answer rewards that conformity cannot raise and message credit set by an impartial judge that re-runs evidence.

Why it will succeed. Each task extends an established principle to agents that share mistakes, argue, and may collude (Table 2). Most tasks build on the PI's previous work: PRD and the PeerEval leaderboard (Li et al., 2024a), FG-PRM (Li et al., 2025), HiPRAG (Wu et al., 2026), and MEQA (Li et al., 2024b) (Section 4).

2 Technical Approach

2.1 How the Problem Is Solved Today, and the Limitations

Six lines of work bear on the problem. We build on each and note the gap that remains (Table 1).

AI judges and voting. LLM judges are standard for evaluating models and training them with AI feedback (Zheng et al., 2023; Bai et al., 2022; Lee et al., 2024). Panels of judges (Verga et al., 2024) and majority votes over sampled answers (Wang et al., 2023) reduce single-judge biases, but they count every vote equally, so correlated judges add little: when two models are both wrong, they give the same wrong answer about 60% of the time on one leaderboard (Kim et al., 2025).

Estimating reliability without an answer key. For two answers and independent raters of known accuracy, the error-minimizing rule weights each vote by the rater's log-odds of being right (Nitzan and Paroush, 1982; Shapley and Grofman, 1984). Dawid and Skene (1979) estimate the accuracies by expectation–maximization (EM), and later work adds item difficulty (Whitehill et al., 2009), provably good starting points (Zhang et al., 2016), dependence between classifiers (Jaffe et al., 2016), and detection of copying sources (Dong et al., 2009). Recent work models dependence

among LLM judges with a shared latent factor (Balasubramanian et al., 2026) or shared confounders (Zhao et al., 2026). These methods treat raters as passive, never arguing or gaming the estimate, and do not say which questions need more than a vote.

Computational social choice. Epistemic social choice asks which voting rules recover a true answer from independent noisy votes (Caragiannis et al., 2016). Impartial selection ensures that a participant's ratings cannot change its own chance of selection (Holzman and Moulin, 2013), by having groups rate only outsiders (Alon et al., 2011; Aziz et al., 2016) or by randomizing among candidates whose ratings could change the outcome (Kurokawa et al., 2015; Kahng et al., 2018); fair lotteries select panels (Flanigan et al., 2021), and peer prediction scores agreement with peers beyond chance (Dasgupta and Ghosh, 2013; Shnayder et al., 2016). Lev et al. (2023) add reliability weights, but they reward agreeing with the majority, which fails once accurate reviewers are a minority. In our setting, some claims can be verified, impartial mechanisms only choose who argues and who judges while debate decides the answer, and participants can persuade and collude while the decision is made.

Discussion and debate among LLMs. Most multi-agent LLM systems use open discussion, revising until agents agree (Du et al., 2024; Chan et al., 2024; Li et al., 2024a). Prior work on LLM discussion is limited: majority voting explains most of its gain (Choi et al., 2025), one round can make judges nearly unanimous without making them more accurate (Shu, 2026), and it assumes honest agents that exchange only text. Standard debate (Irving et al., 2018) assigns two debaters opposite answers and helps judges who know less than the debaters (Michael et al., 2023; Kenton et al., 2024): with quotations verified against the source, non-expert AI judges reached 76% accuracy and human judges 88%, against 48% and 60% without debate (Khan et al., 2024). But it trusts the judge, assumes the sides truly compete, and grounds claims at most in quoted text. Recent systems discuss only when agents are unsure (Eo et al., 2025), though in one study most items where discussion helped were unanimous but wrong (Hu et al., 2026b), or give agents tools (Jeong et al., 2026) or a tool-using verifier for deadlocks (Li et al., 2026b), but no judge re-runs the agents' own calls; judges shown a log claiming that tool calls succeeded accepted most failed outputs (Xu, 2026). We know of no debate protocol designed for many agents that share mistakes or collude.

AI control and adversarial crowdsourcing. AI control keeps protocols safe even if a model subverts them, using a small budget of trusted audits and monitors validated on planted examples (Greenblatt et al., 2024), and in adversarial crowdsourcing a number of trusted ratings that does not grow with the number of items suffices even when honest raters are a minority (Steinhardt et al., 2016). But colluding workers learn to recognize gold questions (Checco et al., 2018), and capable LLMs can often tell an evaluation from real use (Needham et al., 2025), which motivates our two kinds of audits.

Self-improvement without labels. Models improve without labels by training on answers most samples agree on (Huang et al., 2023; Zuo et al., 2025), on multi-agent discussion (Subramaniam et al., 2025; Park et al., 2025), or on debate self-play (Arnesen et al., 2024), but the reward is the weak point. Rewarding majority agreement collapsed accuracy (Shafayat et al., 2025), and label-free intrinsic rewards tend to rise and then collapse (He et al., 2026); rewarding debate wins held off reward hacking, but with a weaker judge the critic learned to fabricate quotes and flaws (Kenton et al., 2026); and rewarding approval taught models to mislead evaluators (Wen et al., 2025). Peer prediction trains LLMs toward truthfulness without labels, but its guarantees exclude

Table 1: How current approaches address the four failures: ● addressed, ◐ partly, ○ not addressed; †where a tool applies. For this project, the marks show what the design targets (Section 2.7).

	Counts shared mistakes once (F1)	Resists strategic or colluding agents (F2)	Evidence the judge can re-run (F3)	Improves agents without labels (F4)	Many agents of unknown skill
Single AI judge (Zheng et al., 2023)	○	○	○	◐	○
Majority vote, juries (Verga et al., 2024; Wang et al., 2023)	○	○	○	◐	◐
Rater models without an answer key (Dawid and Skene, 1979; Dong et al., 2009)	◐	○	○	○	●
Impartial peer review and panel selection (Kurokawa et al., 2015; Flanigan et al., 2021; Lev et al., 2023)	○	◐	○	○	◐
Open discussion among LLMs (Du et al., 2024; Chan et al., 2024)	○	○	○	◐	◐
Standard debate (Irving et al., 2018; Khan et al., 2024)	○	◐	◐	◐	○
AI control (Greenblatt et al., 2024)	○	●	○	○	○
This project	●	●	●†	●	●

collusion (Qiu et al., 2026), and rewarding agreement with other models' majority does not correct for chance agreement and relies on model diversity against shared mistakes (Yang et al., 2026). Process rewards score one agent's steps (Lightman et al., 2024; Li et al., 2025), and debate self-play credits each speech toward its own side (Arnesen et al., 2024), not toward the group's decision.

2.2 Problem Setting and Design Rules

Agents and questions. A pool of AI agents, each built on an underlying model (which several may share) with tools and role instructions, handles a stream of questions. A question q asks for a decision or an evaluation with a small set of candidate answers $\mathcal{Y}$, such as yes or no, a choice among options, or a score band for a paper. For each question, the harness assigns at random which agents answer and which are held out as possible judges, so no agent can become a judge by declining to answer. Each answering agent first answers alone, giving an answer x_{aq}, a step-by-step argument, and its evidence.

Unknowns, resources, and threat model. The true answer y_q, each agent's skill, which agents share mistakes, and which are dishonest are unknown. Available are planted cases (questions whose answers we know, mixed into the stream), a small, fixed budget of human audits (a person or a direct measurement finds the right answer for decisions drawn at random), and tools whose results can be re-run in a sandbox. Dishonest agents may coordinate and may answer, rate, argue, or judge strategically; they know the protocol but not its random draws, and may try to recognize planted cases. Honest agents may still share mistakes, for example from similar training data.

Output and design rules. Each decision comes with its reasons: whose answers carried weight,

which step decided the debate, and which evidence the judge reproduced. Five rules hold throughout. (R1) Every answering agent answers alone, before seeing the others. (R2) Peer ratings pick only who argues; the weighted vote (for a clear leader) or the judge (after a debate) picks the answer. (R3) Tool evidence counts only if the judge re-runs it and obtains the same result. (R4) Audits only catch collusion; they never settle a debate, and nothing triggers a human audit, so its cost stays fixed. (R5) Planted cases never enter training, so agents cannot learn to spot them.

2.3 Formal Framework: Where a Group's Errors Come From

Each task controls one source of error in the group's decision. Let π_q be the posterior probability of the leading answer $\hat{y}_q$ from Task 1.1, and let the question go to debate when $\pi_q < \tau$. The decision D_q is $\hat{y}_q$ without a debate and the judge's choice after one, and its chance of being wrong splits into terms that the tasks control:

$$P(D_q \neq y_q) \;\leq\; \underbrace{(1-\tau)\,P(\pi_q \geq \tau)}_{\text{no debate: Task 1.1}} + P(\pi_q < \tau)\,\Big[\; \underbrace{\varepsilon_{\text{cand}}}_{\text{Task 1.1}} + \underbrace{\varepsilon_{\text{role}}}_{\text{Tasks 1.2, 2.2}} + \underbrace{\varepsilon_{\text{step}}}_{\text{Task 2.1}} \;\Big]. \tag{1}$$

The first term holds when the posterior is calibrated (answers given probability π are right a fraction π of the time); a vote that ignores shared mistakes is overconfident and breaks it (Figure 2(a)), so Task 1.1 supplies a posterior that accounts for them and sets τ by weighing a debate's cost against the stakes. In a debate, $\varepsilon_{\text{cand}}$ is the chance that neither debated answer is right, $\varepsilon_{\text{role}}$ the chance that a dishonest agent holds a role and steers the outcome, and $\varepsilon_{\text{step}}$ the chance that a debate without such steering still ends wrong. Task 1.2 keeps any agent's ratings on a question from raising its own chance of a role on it and Task 2.2 flags colluders, so $\varepsilon_{\text{role}}$ shrinks as audits accumulate; Task 2.1 reduces the judge's task to single disputed steps whose evidence it re-runs, so wherever a tool applies, the judge's share of $\varepsilon_{\text{step}}$ approaches the tools' error rate; and Task 3 lowers the agents' own error rates, which can shrink every term. Removing a task leaves its term uncontrolled, and our ablations measure each term (Section 2.7).

2.4 RG1: Decide What to Debate, and Pick Debaters and Judges Impartially

A plain vote lets shared mistakes pile up, and agents choosing debaters can favor themselves. RG1 addresses both before any debate begins.

2.4.1 Task 1.1: Deciding what to debate, counting shared mistakes once

Challenge. A vote is informative only to the extent that voters err independently, and correlated votes can void the jury theorem's guarantee (Ladha, 1992). In Figure 2(a), five independent agents, each right 70% of the time, reach 84% by majority vote, but agents copying one source of the same accuracy drag the vote to 70%, and a rater model that assumes independence falls sooner, reading the copies' agreement as skill. LLM errors are correlated in practice (Kim et al., 2025; Goel et al., 2025; Kohli, 2026).

Model and method. Each answer counts according to its agent's track record, estimated from who agrees with whom across real questions, as one grades an exam without the answer key. We extend the rater model of Dawid and Skene (1979) with shared error factors; unlike recent dependence-aware aggregators for LLM judges (Balasubramanian et al., 2026; Zhao et al., 2026), a group shares a wrong answer, not a label-independent tilt or style effect, its weight is bounded (Proposition 1), and the posterior decides what to debate. Agents are partitioned into groups g,

Table 2: The principle behind each task, the property it gives the framework, the hypothesis that experiments can refute, and where the property fails. Propositions 1–6 are proved in Sections 2.4–2.6.

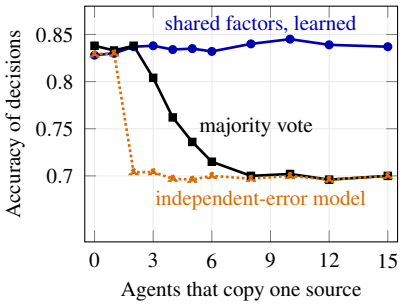
Task	Principle	Property	Hypothesis	Fails when
1.1 Weigh (F1)	Bayes' rule with shared error factors (Dawid and Skene, 1979)	A group counts at most as one agent erring at the group's shared rate (Prop. 1); learnable without an answer key from three or more independent groups (Prop. 2)	H1: a gain over majority vote that grows as mistakes are more widely shared	fewer than three groups err independently
1.2 Choose roles (F2)	Impartial selection by partition (Alon et al., 2011)	No group's ratings change its members' chances of a role; a representative's quality is at least π_y^* times the best (Prop. 3); f colluders judge with probability at most $f\bar{p}$	H2: no gain for a lone strategic rater; impartiality costs less than debate adds	colluders spread across groups; few model families
2.1 Debate (F3)	Debate reduces a question to single steps (Irving et al., 2018)	The judge errs only if it misrules a disputed step, or the right side argues a false step or finds none (Prop. 4)	H3: beats open discussion, more so as more steps can be settled by a tool	hidden flaws; no tool applies; selective facts
2.2 Audit (F2)	Random inspection (Avenhaus et al., 2002; Ramdas et al., 2023)	Every steered decision is audited with probability ρ (Prop. 5); agents erring as predicted are flagged with probability below α	H4: colluders cheating at a fixed rate are flagged after a bounded number of audits	a group that cheats once or rarely
3 Learn (F4)	Noisy labels and peer prediction (Dasgupta and Ghosh, 2013)	Expected update is $1-2\eta_q$ times the update from true answers; answers that ignore the question earn nothing (Prop. 6)	H5: larger gains than rewards for agreement or debate wins, without more dishonesty	references wrong more often than right; groups that Task 1.1 misses

most with one member. On each question, a group has a hidden shared answer s, such as what an outdated schedule says: the true answer with probability $1-\lambda_g$, and a wrong answer w with probability $\lambda_g \nu_g(w \mid y_q)$. Each member a reports s with its own errors, given by its confusion matrix θ_a (the chance of each answer it gives for each value of s). With $\kappa_g(s \mid y) = 1-\lambda_g$ if $s = y$ and $\lambda_g \nu_g(s \mid y)$ otherwise,

$$P(x_{\cdot q} \mid y_q) = \prod_g G_{gq}(y_q), \qquad G_{gq}(y) = \sum_{s \in \mathcal{Y}} \kappa_g(s \mid y) \prod_{a \in g_q} \theta_a(s, x_{aq}), \tag{2}$$

where g_q is the set of members of g that answered q; for one-member groups $\lambda_g = 0$, which is the original model. Decisions use Bayes' rule, the posterior over y_q given all answers; with independent agents, two answers, and equal prior odds, this is the classical error-minimizing vote, in which an agent right with probability p_a has weight $\log \frac{p_a}{1-p_a}$ (Nitzan and Paroush, 1982; Shapley and Grofman, 1984). Parameters are fitted by EM over all real questions, with audit flags as a prior that part of an agent's answers are adversarial; candidate partitions come from clustering agents by agreement, each fitted from a vote in which each candidate group counts once, and the one with the best penalized likelihood is kept. Shared sources serve only as hints, since agents can read the same report and reason differently. In the vessel example, the two agents with the outdated schedule have often given the same unusual answers (Figure 2(b)), so together they count about once, and with no clear leader the question goes to debate.

Why this method. *Proposition 1 (a group counts at most once).* For yes-or-no questions, a group with $\lambda_g \le \frac{1}{2}$ changes the log-odds of the answer by at most $\log \frac{1-\lambda_g}{\lambda_g}$, whatever its size and whatever

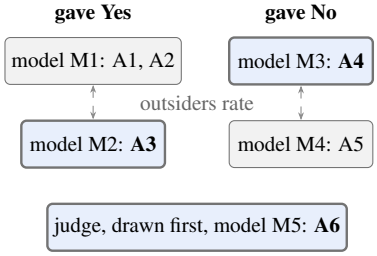


(a) Copies drag a vote down; a group counts once.

	q_1	q_2	q_3	q_4	q_5	q_6
shared factor A1	a	c	a	c	d	a
shared factor A2	a	c	a	c	d	a
A3	a	b	b	c	b	a
A4	a	b	a	c	b	c
A5	b	b	a	c	b	a

A1 and A2 share unusual answers that no one else gives (shaded), so the model infers a shared error factor and counts their agreement about once. The true answers are never needed.

(b) Task 1.1: who agrees with whom.



Dashed arrows: who rates whom. The judge, drawn first from agents that did not answer, rates no one. For each answer, a group without the judge's model is drawn at random, and its top-rated member argues.

(c) Task 1.2: impartial roles.

Figure 2: RG1. (a) Simulation: five independent agents, each right 70% of the time, plus agents copying one source that is wrong 30% of the time (each copy adds 5% errors of its own); means over 20 runs of 400 questions (standard errors below 0.005). Majority vote falls to the source's accuracy, and a rater model assuming independent errors falls once two agents copy. The shared-factor model, fitted without the true answers, recovers the copying group (exactly, in 54 of 60 runs inspected) and, as Propositions 1 and 2 predict, keeps its accuracy. With two or more copies, the independent-error model gives probability 0.95 or more to over 90% of decisions, 28–30% of them wrong, while the shared-factor posterior stays calibrated (calibration error below 0.03). In (c), bold boxes are the agents chosen in the running example.

its members answer. *Proof.* Let r be the ratio of the probabilities of the members' answers when s is yes and when it is no. By (2), the group's likelihood ratio is $\big[(1-\lambda_g)r+\lambda_g\big]/\big[(1-\lambda_g)+\lambda_g r\big]$, which increases with r and lies between $\frac{\lambda_g}{1-\lambda_g}$ and $\frac{1-\lambda_g}{\lambda_g}$. □

So a group carries at most the weight of one agent that errs with probability λ_g, whereas under independence each member would multiply the ratio again; with more answers and wrong answers equally likely, the bound becomes $\log\big[(|\mathscr{Y}|-1)\frac{1-\lambda_g}{\lambda_g}\big]$.

Proposition 2 (learning without an answer key). Fix the partition, and suppose every answer has positive prior probability, at least three groups answer each question, their answers are independent given the true answer, and each group's answer distributions under different true answers are linearly independent (as when its members' confusion matrices are invertible and $\lambda_g < \frac{1}{2}$). Then each group's distribution of answers given the true answer, which is all the posterior uses, is identifiable from the answers alone, up to relabeling, by Kruskal's theorem on three-way arrays (Kruskal, 1977; Allman et al., 2009). With only two such groups it generally is not: different track records explain the same answers equally well, so no method that sees only answers can tell an accurate majority from one that copies a mistake.

Proposition 2 states the method's limit in advance, and our experiments include pools where it binds; Figure 2(a) shows that the model can be learned in practice.

Deciding what to debate. With a calibrated posterior, letting the leading answer stand risks a loss L with probability $1-\pi_q$, while a debate costs c and, for two answers, ends wrong with a fixed probability ε_D. A debate is worth its cost exactly when $L(1-\pi_q)-L\varepsilon_D > c$, that is, when

$$\pi_q < \tau = 1-\varepsilon_D - c/L, \tag{3}$$

so higher stakes lead to more debates. We estimate ε_D on development questions with known

answers, separately for disputes a tool can settle. Because π_q counts effective votes, a unanimous answer from agents that share a mistake can still fall below τ and go to debate; rules triggered by low confidence or disagreement (Eo et al., 2025) miss such questions, which were most of those where discussion helped in one study (Hu et al., 2026b). Weighting cannot restore the information a correlated panel lacks (Kohli, 2026), so deciding what to debate is the posterior's main job.

Research directions. (a) *Detecting groups.* A copying group is exposed only where it gives an answer others do not. We will characterize how many questions it takes to detect a group, as a function of its size, λ_g, and how rarely others give its wrong answers, and when the partition itself is identifiable. (b) *Topics, difficulty, and calibration.* Topic-specific confusion matrices and question difficulty (Whitehill et al., 2009) keep hard questions from being read as copying, and the posterior is recalibrated on development questions.

Our previous work. PRD's Peer Rank, which weights each AI reviewer by how highly the others rate its answers, agreed with human preferences more often than the strongest single judge (67.3% vs. 64.3% for GPT-4) (Li et al., 2024a); since such weights come from peers' ratings, models that share mistakes can raise one another's standing, which track records prevent.

Evaluation and risks. On testbed A, where some agents receive the same flawed hint, we measure recovery of the seeded groups, accuracy, calibration, and questions needed to detect a group; plain Dawid–Skene is an ablation. H1 predicts a gain over majority vote that grows with how widely mistakes are shared and disappears with fewer than three independent groups. If EM converges poorly, we will start it from a spectral method (Zhang et al., 2016) or use a pairwise measure of agreement on mistakes (Goel et al., 2025).

2.4.2 Task 1.2: Impartial choice of debaters and judges

Challenge. A debate needs a judge and one representative per competing answer. If agents choose them by rating one another, an agent can rate rivals low to be picked, agents built on similar models favor one another's outputs (Panickssery et al., 2024; Goel et al., 2025), and a judge that shares a model with one side has a stake in the outcome.

Core method. Representatives are chosen by peer ratings of the arguments behind first answers, with raters weighted by track record, building on PRD and PeerEval (Li et al., 2024a); we adapt the partition idea of impartial selection (Alon et al., 2011; Aziz et al., 2016). Agents sharing an underlying model form a group, fixed before any rating; $g(a)$ is agent a's group, H_q the agents that did not answer q, and E_y the groups with a member answering y. On question q: (1) the judge $J \sim \mathrm{Unif}(H_q)$ is drawn first and rates no one; (2) for each debated answer y, a group K_y is drawn from $E_y \setminus \{g(J)\}$ with probabilities π_y that do not depend on ratings, avoiding groups drawn for other answers where possible; (3) its representative is

$$R_y = \operatorname*{arg\,max}_{a \in K_y : x_{aq} = y} \sum_{b \in O_y(a)} w_b \, r_{ba}, \tag{4}$$

where r_{ba} is b's rating of a's argument, w_b is b's track-record weight (Task 1.1), and the raters $O_y(a)$ are the agents outside $g(a)$, other than J, that answered y or did not answer. An answer with no eligible group, such as the alternative to a unanimous answer sent to debate, gets a random agent outside $g(J)$, as in standard debate. Ratings pick only who argues; the judge picks the answer (Figure 2(c)).

Why this method. *Proposition 3 (impartiality and its price).* (a) For any group g and any two rating profiles that differ only in ratings given by members of g, each member of g has the same probability of being the judge or a representative. (b) If ratings are honest and rank each group's members by argument quality $Q \geq 0$, then $\mathbb{E}[Q(R_y)] \geq \pi_y^* \, Q(a_y^*)$, where a_y^* is the best agent answering y and π_y^* the probability that its group is drawn. *Proof.* (a) J, the debated answers, and every K_y are drawn independently of ratings. If $K_y \neq g$, no member of g represents y; if $K_y = g$, (4) uses no rating from g, and the weights do not depend on ratings. (b) Whenever $g(a_y^*)$ is drawn, (4) selects a_y^*. □

A single agent is a one-member coalition, so (a) gives impartiality in the sense of Kurokawa et al. (2015), extended to collusion within a group. With uniform draws, (b) gives $1/|E_y|$ unless the judge's group holds a_y^*; with two or three groups per answer, this beats the $1/(1+m)$ guaranteed by randomizing over would-be winners (Kurokawa et al., 2015) when one agent is chosen and each rates m others. No answering agent rates the other side and the judge rates no one, so neither can rate the other side's best arguer low. A judge lottery that caps any agent's chance at $\bar{p}$ lets f colluders hold the seat with probability at most $f\bar{p}$, so judge chances are kept as equal as possible (Flanigan et al., 2021). Colluders in other groups, or held out, can still shift ratings, so we measure coalition gains.

Research directions. (a) Agents that share error factors, or recognize one another's outputs (Panickssery et al., 2024), will share a group even when their model labels differ. (b) Drawing groups in proportion to a score that ratings cannot change, such as the best track record in the group, keeps Proposition 3(a) and raises π_y^* when that score tracks quality; we will measure the price of impartiality and bound what coalitions gain.

Our previous work. In PRD, with GPT-4 as the only reviewer, GPT-4's Elo rating (a chess-style rating) was 46 points above the one derived from human preferences, while weighted peer ranking placed all five models in the human order (Li et al., 2024a).

Evaluation and risks. We measure manipulability, the largest gain in selection probability a strategic rater or coalition achieves by misreporting (found by the red team), the price of impartiality, and the accuracy of the resulting debates (H2), on PeerEval and testbed C. With few underlying models there are few groups; we will then diversify agents by role instructions and tools, or draw judges from a held-out pool of models that never answer.

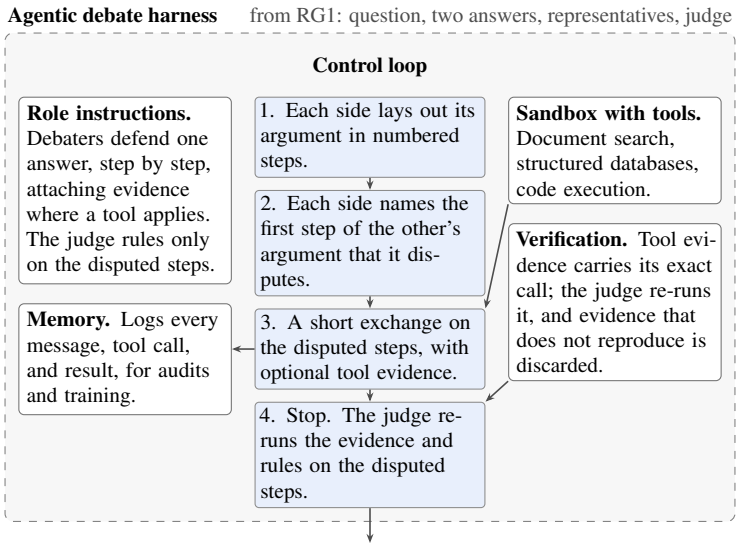
2.5 RG2: Settle Disputes by Agentic Debate, and Catch Collusion

Open discussion erases disagreement and rewards persuasion, and debaters can collude. RG2 settles each dispute by debate before an impartial judge and audits the participants.

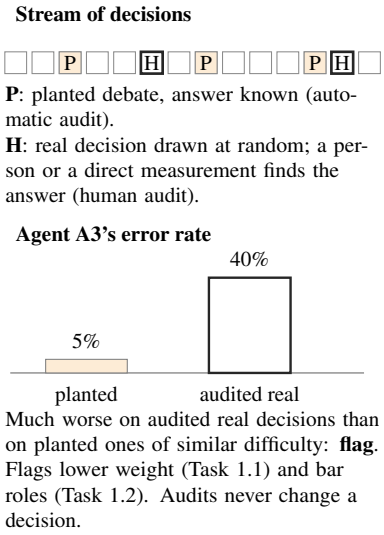
2.5.1 Task 2.1: Debate before an impartial judge

Challenge. Today's multi-agent systems mostly discuss until they agree, which erases the disagreement that exposes errors (Shu, 2026; Choi et al., 2025). Standard debate keeps the disagreement, but it runs fixed rounds of text between given models and leaves the judge to weigh words alone, so a persuasive debater can win a step that one database query would settle.

Core method. We run debate in a harness, the software that turns models into agents (Ning et al., 2026; Yao et al., 2023; Yang et al., 2024), with five parts (Figure 3(a)). *Role instructions* tell each debater which answer to defend, to argue in numbered steps, and to attach evidence where a tool applies, and tell the judge to rule only on disputed steps. The *control loop* sets who speaks and



(a) Task 2.1: debate before an impartial judge.



(b) Task 2.2: two kinds of audits.

Figure 3: RG2. (a) The control loop sets who speaks, who judges, and when to stop; the judge rules on the disputed steps after re-running any tool evidence (Proposition 4). (b) Automatic audits mix planted debates into the stream, and human audits cover a few real decisions drawn at random; an agent that behaves well only on planted debates is exposed by the gap between its two error rates (illustrative numbers).

who judges (picked in Task 1.2) and when to stop: the representatives lay out their arguments in numbered steps, each names the first step of the other's argument that it disputes, a short exchange follows, and the loop stops for the judge, who upholds or strikes each disputed step (Irving et al., 2018). *Tools*, such as document search, databases, and code execution, are optional. *Verification*: tool evidence carries its exact call (tool, arguments, data version), which the judge, itself an agent (Zhuge et al., 2025), re-runs; evidence that does not reproduce is discarded. *Memory* logs every message, call, and result. In the vessel example, A4 disputes A3's step "so V left at 06:00" with a 09:10 port report listing V at its berth, and the judge re-runs A4's query and picks no (Figure 1).

Why this method. *Proposition 4 (judging a question reduces to judging single steps).* Suppose that (i) every argument whose steps are all true reaches the right answer, (ii) the side defending the right answer argues with true steps and can find a false step in the other side's argument, and (iii) the judge rules correctly on any single step it is shown. Then, if a side loses exactly when one of its disputed steps is struck, the side defending the right answer wins. *Proof.* The other side's argument reaches a wrong answer, so by (i) it contains a false step, which the right side disputes and the judge strikes; any step of the right side's argument that the other side disputes is true, and the judge upholds it. □

So the judge errs on a debated question only when (ii) or (iii) fails on a disputed step, the core idea of debate (Irving et al., 2018; Brown-Cohen et al., 2024). Re-running tool evidence makes (iii) plausible: a step a tool can settle is ruled on about as accurately as the tool itself, so the judge's share of $\varepsilon_{\text{step}}$ in (1) approaches the tool's error rate, while the judge examines a few steps instead of two whole arguments. The guarantee fails where its assumptions do: a flaw hidden among many steps that are each hard to verify, the obfuscated-argument problem (Brown-Cohen et al., 2025); steps no tool can settle; and true but selectively chosen facts (Hu et al., 2026a).

Research directions. (a) *Finding the false step.* We will propose candidate false steps with models that judge whether one claim implies another and with FG-PRM's step-level error detection (Li et al., 2025), and recurse into a complex disputed step's sub-steps, following prover–estimator debate (Brown-Cohen et al., 2025). (b) *A right of reply.* Before the judge rules, each side may add context to the other side's evidence, against selectively chosen facts, and quotations are verified against the source (Khan et al., 2024). Ablations remove each harness part, remove tools or re-running, and replace debate with open discussion among the same agents.

Our previous work. In PRD's peer discussion, agreement with human preferences rose from 72.9% (GPT-4 alone) to 75.0% when GPT-3.5 led the discussion, but fell to 71.4% when GPT-4 led (Li et al., 2024a): one reviewer can carry a discussion. HiPRAG's search agents, which ground answers in retrieved evidence (Wu et al., 2026), will be our first tool-using debaters.

Evaluation and risks. We measure accuracy on disputed questions, correct minorities that prevail and correct majorities overturned, how often re-running evidence changes the outcome, the judge's error on each kind of step, and cost. H3 predicts that debate beats open discussion among the same agents, more so as the share of steps a tool can settle grows. If neither side finds a false step, the judge reads both arguments, as in standard debate, and a close ruling is reported as unsettled, with its probability and the step in dispute; tool evidence can be out of date, so the judge weighs its provenance, and human audits measure how often this happens.

2.5.2 Task 2.2: Automatic and human audits against collusion

Challenge. Collusion means debaters, or the judge, steering a debate to the wrong side: arguing badly on purpose (van der Weij et al., 2025), misleading with true but selectively chosen facts (Hu et al., 2026a), or favoring one side, none of which re-running evidence catches. LLM agents collude when it pays (Shi et al., 2026), can hide signals in ordinary text (Mathew et al., 2025), and can often tell evaluations from real use (Needham et al., 2025).

Core method. We use two kinds of audits (Figure 3(b)). *Automatic audits* mix planted debates on planted cases into the real stream, formatted like real questions; a planted case can be a paper with a deliberately inserted error, as in studies of human peer review (Schroter et al., 2008). *Human audits* cover a few real decisions drawn at random after they are made: a person or a direct measurement, such as a satellite image, finds the right answer. Blame is assigned by comparing each agent with the others it was paired with across many random pairings. An agent whose audited decisions go wrong more often than its track record predicts, or that does much worse on audited real decisions than on planted ones of matched difficulty, is flagged; flags lower its weight in Task 1.1 and bar it from arguing or judging. Audits never change a decision, and nothing triggers a human audit.

Why this method. *Proposition 5 (random audits cannot be dodged).* If each real decision is audited independently with probability ρ, drawn after the decision is made, then every decision a colluder steers is audited with probability ρ, whatever the colluder can infer about planted cases, and the chance that its first n steered decisions all escape audit is $(1-\rho)^n$, however long the stream. *Proof.* The draw is independent of everything the colluder saw when it acted. □

Planted cases add cheap evidence as long as they cannot be told from real questions; human audits keep the guarantee when they can. Random audits deter agents that can hide misconduct only if the evidence survives and the timing is unpredictable (Gans and Holden, 2026; Allingham and Sandmo, 1972); immutable logs and after-the-fact draws provide both. Flags are raised by an anytime-valid

sequential rule (Ramdas et al., 2023), which keeps the chance that an agent erring no more than predicted is ever flagged below a chosen level α (split across agents) however long monitoring runs; the number of audits needed to flag a colluder then depends on α and on how much more often its decisions go wrong, not on the number of decisions, in line with Steinhardt et al. (2016).

Research directions. (a) *Planted cases that look real*, made from real questions by controlled edits, such as changing a timestamp in a port report, with a discriminator that guides the generator until it can no longer tell planted from real (Checco et al., 2018). (b) *How many audits are enough.* Treating auditing as an inspection game (Avenhaus et al., 2002), we expect that a colluder that tunes how often it cheats can cause, before it is flagged, wrong decisions, beyond the honest error rate, that grow at most like the square root of the number of decisions when honest agents also err, and stay bounded when they rarely do; we will prove such bounds and compare them with the red team's best attacks. A colluder that mimics a weak honest agent is never flagged, so its harm is bounded instead by the weight and roles a weak record earns (Task 1.2(b)). (c) *Signs of coordination*, such as rulings that favor one debater more than the evidence explains (Dong et al., 2009), with messages paraphrased against hidden signals (Roger and Greenblatt, 2023).

Our previous work. The PI co-advised UT Dallas's team, one of five university red teams in the Amazon Nova AI Challenge (Sahai et al., 2025), which built COMET, a closed-loop red-teaming framework that elicited harmful outputs from guarded code models (Xu et al., 2025).

Evaluation and risks. Against the red team, we measure wrong decisions that get through at each audit budget, time to flag colluders, false flags, and whether a discriminator can tell planted cases from real ones (H4). A group that cheats only once is caught with probability at most ρ, and if planted cases stay recognizable, flags rest on the human audits alone, which slows flagging.

2.6 RG3: Let Agents Self-Improve with Reinforcement Learning on the Group's Decisions

Self-improvement needs rewards that track correct answers, not votes or debate wins.

2.6.1 Task 3: Rewards for contributions to trustworthy decisions

Challenge. The obvious rewards fail (Section 2.1): majority agreement rewards conformity and can collapse accuracy (Shafayat et al., 2025), debate wins reward persuasion when the judge can be fooled (Kenton et al., 2026), and approval rewards misleading the evaluator (Wen et al., 2025). A group's reward must also say whose messages helped, or training rewards agents that contributed nothing and misses the agent that found the decisive evidence.

Core method. We train the agents with reinforcement learning on the group's decisions, which need no human labels and count shared mistakes once. *First answers.* Agent a's first answer to q is scored against D_q^{-a}, the answer that the posterior of Task 1.1 favors when a's group is left out, minus the agreement of its answer to another question q' with that reference on a third, q'', both on its topic and drawn at random:

$$r_a(q) = \omega_q \mathbf{1}\{x_{aq} = D_q^{-a}\} - \omega_{q''} \mathbf{1}\{x_{aq'} = D_{q''}^{-a}\}, \tag{5}$$

so an agent earns credit for agreeing with the rest of the group beyond chance, the correlated-agreement form of peer prediction (Dasgupta and Ghosh, 2013; Shnayder et al., 2016). Unlike pairwise peer prediction (Qiu et al., 2026) or agreement with other models' majority (Yang et al., 2026), the reference counts shared mistakes once and discounts flagged agents. *Debate messages.*

Message m_t in a debate on q is credited by how far it moves the judge toward the decision D_q:

$$r_t = \omega_q \left[p_J(D_q \mid m_{1:t}) - p_J(D_q \mid m_{1:t-1}) \right], \tag{6}$$

where p_J is the judge's probability of the decision given the debate so far, obtained by re-scoring each prefix of the logged transcript. For two answers, $\omega_q = 1 - 2\hat{\eta}_q$ (with more, $1 - \hat{\eta}_q |\mathscr{Y}|/(|\mathscr{Y}| - 1)$), where $\hat{\eta}_q$ is the estimated chance that the reference is wrong: from the left-out posterior for first answers, and from the judge's probability, calibrated on development questions, for messages. For score bands, exact agreement gives way to the correlated-agreement score (Shnayder et al., 2016). Decisions involving flagged agents are dropped, planted cases never enter training, and training uses standard policy-gradient methods (Shao et al., 2024). In the vessel example (Figure 1), A4's message citing the port report moves the judge most and earns the most credit, and A3's unsupported claims move the judge away from the decision and are penalized, whereas a majority-agreement reward would have rewarded the three yes answers.

Why this method. *Proposition 6 (decisions as noisy labels).* Consider a yes-or-no question, and suppose that, given the true answer, D_q^{-a} is wrong with probability η_q independently of agent a's answer. Then $\mathbb{E}\,\mathbf{1}\{x = D_q^{-a}\} = \eta_q + (1 - 2\eta_q)\,\mathbf{1}\{x = y_q\}$, so in expectation, rewarding agreement with D_q^{-a} gives $1 - 2\eta_q$ times the policy-gradient update that rewarding the true answer would give; the second term of (5) does not depend on x_{aq} and leaves this update unchanged. If a's answers ignore the question, the two terms of (5) have the same expectation over questions, so such answers earn nothing. *Proof.* $P(D_q^{-a} = x)$ is $1 - \eta_q$ if $x = y_q$ and η_q otherwise, and terms that do not vary with the answer do not change the policy gradient. Answers that ignore the question, x_{aq} and $x_{aq'}$, share one distribution and are independent of (ω_q, D_q^{-a}) and $(\omega_{q''}, D_{q''}^{-a})$, which also share one distribution. □

Proposition 6 explains each design choice. Every reference more often right than wrong pushes the agent toward the right answer, and ω_q weights each question by that push (an inverse-noise weight, not the unbiased correction of Natarajan et al., 2013): confident references count fully, near-ties little. Independence fails when the reference includes the agent's own answer, since the agent can then raise its reward by agreeing with itself and agents like it, which is how rewards for majority agreement collapse (Shafayat et al., 2025); leaving out the agent's group restores it under the model of Task 1.1, and subtracting chance agreement removes the reward for answers that ignore the question, such as always saying yes, even when colluders coordinate on them. For debate messages, the credits of a debate sum to ω_q times the change in the judge's probability of the decision, so (6) splits that change among messages, telescoping as potential-based shaping does (Ng et al., 1999) (its potential depends on the decision, so policy invariance does not carry over), and credits each message with its effect given the earlier ones, in the spirit of difference rewards (Wolpert and Tumer, 2001; Agogino and Tumer, 2008). When the judge decides by re-running evidence, the decision does not depend on persuasion, so messages earn credit for moving the judge toward what the evidence shows.

Research directions. (a) *Credit at scale.* We will train a multi-agent process reward model on logged debates to predict each message's credit, with labels such as "introduced re-runnable evidence" and "persuaded without evidence," extending FG-PRM, and compare it with order-free credit such as Shapley values. (b) *What improves.* We will measure whether training improves first answers, debating, and judging on held-out questions, topics, and model families, including where human judges are unreliable (Burns et al., 2024). Agents are rewarded separately, agents that judge

Table 3: Testbeds. Each provides known answers, controlled shared mistakes, and tools whose results the judge can re-run.

Testbed	Questions and sources	Shared mistakes and dishonesty	Tools in the sandbox
A. Math and code	GSM8K and MATH problems (Cobbe et al., 2021; Hendrycks et al., 2021); programming problems with reference outputs	Some agents receive the same flawed hint, so the groups that share mistakes are known; dishonest agents	Calculator, code execution
B. Multi-document questions	Questions composed from event structures by our MEQA pipeline (Li et al., 2024b), including synthetic maritime scenarios whose reports form source trees of originals, copies, paraphrases, and outdated versions	Circular reporting built into the source trees; forged reports and spoofed position tracks; a compromised agent	Document search; a port and vessel database built from the same events
C. Scientific claims and peer review	Whether a paper's evidence supports its claims, on public papers with their OpenReview reviews and rebuttals, and the PeerRead corpus (Kang et al., 2018)	Planted errors in papers (Schroter et al., 2008); AI reviewers that share an underlying model	Paper search; code execution for released code

each other never share a reward, and no judge is trained on its own rulings (Shi et al., 2026).

Our previous work. FG-PRM labels each step of a math solution with one of six hallucination types and, as a verifier, substantially improves accuracy on GSM8K and MATH (Li et al., 2025); HiPRAG rewards each search decision of an agent, not only its final answer (Wu et al., 2026).

Evaluation and risks. On all three testbeds, we compare our rewards with rewards for majority agreement and for debate wins (Arnesen et al., 2024) at matched compute, on open-weight models from several families, measuring the accuracy of decisions and first answers on held-out questions and dishonest behavior during training: invented evidence, unsupported changes of answer, and wrong decisions the red team gets through (H5). A misconception most agents share makes η_q exceed one half on some questions, and peer prediction can pay for it (Denisov-Blanch et al., 2026); accuracy on held-out questions with known answers detects this, and training stops if it falls. Training on the same decisions can make agents alike, so we measure shared error factors before and after training and train agents on different subsets of decisions. Debaters could learn to join the judge's likely decision instead of defending their answer; if so, we will credit messages for moving the judge toward the debater's own answer, which makes (6) a shaped debate-win reward.

2.7 Testbeds, Red Team, and Evaluation Protocol

All data are public or synthetic, and every question has a known answer for scoring, kept apart from the planted cases agents may meet. In experiments, a human audit reveals the known answer of a decision drawn at random; we also study audits that are sometimes wrong (Table 3).

A peer-review example. In testbed C, two AI reviewers disagree on whether a paper's experiments

support its claimed five-point gain; the doubter disputes the step "the baseline was evaluated on the same data split," and the judge re-runs the released evaluation script for both settings, so an area chair sees the disputed step and the reproduced numbers.

Agent pools and red team. Agents are built on open-weight models of several families and sizes, plus commercial models where licenses allow, with varied role instructions and tools. In testbed C, published reproducibility reports supply known answers for human audits (Pineau et al., 2021), and agents are not told which claims have reports, so Proposition 5 holds within that set. Following AI control (Greenblatt et al., 2024), a red team builds attackers of increasing strength: (1) a fixed dishonest agent; (2) an agent that knows the protocol but not its random draws; (3) an agent trained to spot planted debates that cheats rarely, and only when it expects no audit; and (4) a debater and a judge that coordinate through hidden signals. We report wrong decisions that get through at each audit budget, together with how much useful work the framework still does, as safety–usefulness curves.

Conditions varied. Each hypothesis predicts where its method should help and where it should fail, so experiments vary the conditions that decide this: (i) how widely mistakes are shared, from no agents to most given the same flawed hint or outdated report, and the number of model families, from one to five; (ii) dishonesty, from none to 30% of agents, at red-team levels 1–4; (iii) evidence, the share of disputed steps a tool can settle and of reports that are outdated or forged; (iv) budgets for debates, audits, and computation, including small models and few tool calls, as under limited connectivity; and (v) answer types, from yes or no to score bands. For example, H1 predicts a gain over majority vote that grows with sharing and disappears with fewer than three independent groups (Proposition 2), and H3 an advantage that grows with the share of steps a tool can settle (Proposition 4). Regimes where the propositions say the methods fail are included, so failures are measured, not assumed; transfer is measured on held-out topics and model families.

Metrics, baselines, and statistics. Primary metrics are decision accuracy and calibration, wrong decisions under attack, correct minorities recovered and correct majorities overturned, audits needed to flag colluders, and cost in computation and tool calls. Each term of (1) is measured separately, so a gain can be traced to the task that produced it. Baselines are a single AI judge, majority vote, and averaging with equal weights, at matched compute; ablations remove each component in turn. We report means with confidence intervals over seeds and question samples.

2.8 Expected Results, New Knowledge, and Limitations

Results and new knowledge. If successful, the project will produce (i) a rater model that counts a group sharing a mistake at most once, with conditions for learning it without an answer key and rates for detecting groups; (ii) an impartial mechanism for choosing debaters and judges, with its measured price and bounds on coalition gains; (iii) an open agentic debate harness that reduces judging a question to judging single steps, with the judge's error measured for each kind of step; (iv) an audit scheme with bounds on the harm a colluder can do before it is flagged; and (v) a self-improvement method whose expected update follows accuracy rather than agreement. Scientifically, it will show when a debate among AI agents can be trusted and how few audits keep the agents honest; because people audit only a few decisions and training needs no labels, the methods are built to scale to AI whose answers people cannot easily evaluate.

Limitations. Where no tool applies, the judge rules on argument alone. A misconception every

agent shares cannot be detected from answers alone (Proposition 2); a group that cheats once is caught at most with the audit probability (Proposition 5); paraphrasing limits but does not remove hidden signals (Mathew et al., 2025); and Navy use would need Navy data and expert validation.

2.9 Future Naval Relevance

ONR S&T Focus Area. Primary: Artificial Intelligence and Autonomy (Intelligent and Collaborative Agents; Adversarial AI & Autonomy). Secondary: C5ISR and Naval Space (Intelligence, Decision Superiority, Command and Control (C2), and Combat Systems). The 2026 ONR S&T Strategy will explore validating AI-produced results by "human review, experimental confirmation or independent AI critique," using AI to "validate, red-team and quality-check" results under human oversight (Office of Naval Research, 2026a); this project supplies methods for such critique when the critics themselves may err together or collude. It also answers the Mathematical Data Science program's call for "rigorous mathematical and algorithmic answers to questions that are currently addressed using heuristics or non-principled approaches" (Office of Naval Research, 2026b).

Navy needs. The problem arises wherever several AI tools report on one question and some reports are copies, stale, forged, or produced by a compromised tool:

- *Maritime domain awareness*: position broadcasts can be spoofed (U.S. Maritime Administration, 2024), and most industrial fishing vessels are not publicly tracked (Paolo et al., 2024), so decisions must combine port records, patrol reports, and imagery.
- *Intelligence analysis*: analysts must describe the "quality and credibility of underlying sources" (ODNI, 2015), and repeated reports from one source can give "an impression of confirming evidence" (WMD Commission, 2005).
- *Decision support by networks of agents*: the Department of War is developing AI agents "for AI-enabled battle management and decision support" (Department of War, 2026), and Marine Corps decision aids seek trust "properly aligned to accuracy of results" (Office of Naval Research, 2026c).
- *Evaluating AI before fielding*: more data and expertise are needed "to quantify performance and risk of AI-enabled systems" (DOT&E, 2026), as AI judges increasingly score other AI systems.
- *Maintenance, logistics, cyber defense, and collaborative autonomy*: conflicting logs and records, alerts triaged by several tools, and unmanned systems that share reports under limited communications.

Across these uses, Task 1.1 counts copied and stale reports once; Tasks 1.2, 2.1, and 2.2 keep a compromised agent from rating its way into a role, discard evidence that does not reproduce, and flag colluders; Task 3 improves agents when ground truth arrives late; and each decision comes with its reasons.

Path to transition. The project is fundamental research. Its products are methods with stated guarantees, an open harness, testbeds, and protocols that Navy laboratories, such as NRL's Navy Center for Applied Research in Artificial Intelligence, could evaluate on Navy data. The PI will pursue ONR Summer Faculty Research Program and Naval Research Enterprise Internship Program placements for students to work with Navy scientists on such evaluations.

Table 4: Schedule by half-year (Y1 = April 2027 – March 2028) and milestones.

Activity	Y1a	Y1b	Y2a	Y2b	Y3a	Y3b
Testbeds A and B; red-team levels 1–2						
Testbed C; red-team level 3						
Task 1.1: shared-factor model, identifiability; then topics, stakes						
Task 1.2: impartial selection; then coalitions, price of impartiality						
Task 2.1: harness v1; then false-step search, right of reply						
Task 2.2: planted and human audits; then audit bounds						
Task 3: rewards (5)–(6); then credit model, comparison						
Red-team level 4; full evaluation; release						

Milestones. M1 (month 6): harness v1 and testbed A released. M2 (12): shared-factor model and its identifiability analysis, on testbeds A and B. M3 (18): impartial selection integrated; tool-backed debate on testbed B. M4 (24): audit scheme and bounds against red-team levels 1–3; testbed C. M5 (30): analysis and comparison of rewards. M6 (36): full framework against level 4; code, testbeds, and protocols released.

Table 5: Goals, benchmarks (testbeds A–C, PeerEval), and success criteria, fixed before the experiments.

Goal and benchmark	Success criterion
G1 (Task 1.1). Accuracy, calibration, and recovery of seeded groups across levels of sharing (A, B)	A margin over majority vote that grows with how widely mistakes are shared, where three or more groups err independently (H1, Year 2)
G2 (Task 1.2). Selection gain of strategic raters and coalitions; price of impartiality (PeerEval, C)	No gain for a single rater; coalition gains below a preset bound; impartiality costs less than debate adds (H2, Year 2)
G3 (Task 2.1). Accuracy on disputed questions; judge's error on each kind of step (B, C)	More accurate than open discussion and all baselines where a tool can settle the disputed steps (H3, Year 2)
G4 (Task 2.2). Wrong decisions that get through, time to flag, false flags, at red-team levels 1–4	Colluders cheating at a fixed rate flagged after a number of audits that does not grow with the stream; false flags below the chosen level (H4, Year 3)
G5 (Task 3). Held-out accuracy of decisions and first answers; rates of dishonest behavior	Larger gains than agreement and debate-win rewards, with no rise in dishonesty (H5, Year 3)
G6. Code, testbeds, protocols; peer-reviewed papers with proofs	Public release by month 36

2.10 Project Schedule and Milestones

The project runs 36 months from April 1, 2027. Table 4 gives the schedule by half-year; milestones M1–M6 mark results.

2.11 Goals and Benchmarks

Each goal has a benchmark and a criterion for success (Table 5); results that miss the criteria will be reported as findings.

2.12 Gold Standard Science

The project will follow the tenets of Executive Order 14303, as the PI's previous work does.

Reproducible and transparent. The harness logs every debate with its exact tool calls, model versions, prompts, and seeds, and tool evidence is re-runnable by design. Code, testbeds, prompts, and protocols will be released with each paper; the PI's group studies reproducibility directly (Yan et al., 2025).

Communicative of error and uncertainty. Every guarantee is stated with its assumptions, decisions carry calibrated probabilities, and the framework's accuracy is estimated from random human audits with valid confidence intervals (Angelopoulos et al., 2023).

Skeptical and falsifiable. A red team attacks our own methods, ablations remove each component, and each task states a hypothesis that experiments can refute (H1–H5, Table 2) and the regime in which its method should fail. Metrics and analysis plans will be fixed before the main experiments.

Collaborative, peer reviewed, and open to negative results. The work joins statistics, social choice, AI safety, and natural language processing, with results for peer-reviewed venues in each; failures of debate or audits will be published as findings.

Without conflicts of interest. The PI will disclose outside activities and support as UT Dallas policy requires, and evaluations use public or synthetic data with released code.

3 Management Approach

This is a single-PI award with no subawardees, co-investigators, or senior personnel. The PI will direct all research and be responsible for schedules, reports, and releases. The budget supports two Ph.D. students for three years, one month of PI summer effort per year, travel to ONR Project and Program Review Meetings and to conferences, and computing.

Team and roles. Student 1 leads the statistical and mechanism-design work and its proofs: Tasks 1.1 and 1.2, the audit analysis of Task 2.2, and testbeds A and C. Student 2 leads the agent and learning work: the harness of Task 2.1, Task 3, and testbed B, which reuses the MEQA pipeline. Both students run the red team, rotating roles so that no one attacks only their own method. Code, logs, and planted-case generators live in one shared repository, with the harness frozen at each milestone so that results stay comparable.

Reporting and dissemination. The PI will submit annual and final reports, attend Project and Program Review Meetings, publish in peer-reviewed venues, and release code, testbeds, and protocols at M1, M6, and with each paper.

Relation to other support. The PI's NSF CAREER award studies event knowledge from long documents, the subject of the MEQA pipeline that testbed B reuses; the Open Philanthropy grant supports control evaluations of two-agent debate, not the audits, selection, and rewards proposed here; the Amazon Research Award concerns process-guided fine-tuning. This proposal funds aims those awards do not, and any overlap with a pending proposal would be removed with the program officers.

4 Principal Investigator Qualifications

Position and training. The PI, Xinya Du, has been an Assistant Professor of Computer Science at The University of Texas at Dallas since 2022, in a first tenure-track appointment. The PI received a Ph.D. in Computer Science from Cornell University in 2021, advised by Claire Cardie, and was a postdoctoral researcher with Heng Ji at the University of Illinois Urbana-Champaign, where the PI was first author of RESIN-11, a schema-guided event prediction system developed in DARPA's KAIROS program (Du et al., 2022). The PI was named a University of Chicago Rising Star in Data Science in 2021.

Recognition and service. NSF CAREER Award (2024–2029); Amazon Research Award (2023); Cisco Research Award (year). The PI's Ph.D. paper on neural question generation (Du et al., 2017) has been cited more than 700 times and is ranked by Paper Digest among the most influential ACL 2017 papers. The PI has served as an area chair for ACL (2023, 2024, 2025), EMNLP (2024, 2025), LREC-COLING 2024, and COLING 2025, and as Website Chair of NAACL 2024, and co-organized the AI4Research workshop at AAAI 2025 and the Workshop on Multimodal Knowledge and Language Modeling at IJCAI 2025.

Independent contributions since the Ph.D. With students at UT Dallas, the PI has built a line of work on AI evaluating AI that this project extends.

- *Peer evaluation among AI judges.* PRD (Li et al., 2024a) introduced peer ranking and peer discussion among AI judges, which agree with human judgments better than the strongest single judge and reduce a judge's bias toward its own model. It underlies PeerEval (peereval-ai.github.io), a public leaderboard on which seven AI models answer the same questions and judge one another anonymously (40,156 pairwise judgments in its first release). IQA-EVAL (Li et al., 2024c) evaluates interactive question answering with persona-driven AI agents.
- *Rewards and verification.* FG-PRM (Li et al., 2025) detects six types of hallucination at each step of a math solution; HiPRAG (Wu et al., 2026) rewards each search decision of an agent, making agentic search more accurate and efficient; and Liang et al. (2026) show that a reasoning model can verify answers without further training.
- *Benchmarks and AI for science.* MEQA (Li et al., 2024b) provides event-centric multi-hop questions with explanations, FaithScore (Jing et al., 2024) measures fine-grained hallucinations, and LMR-Bench (Yan et al., 2025) evaluates agents that reproduce research code; the PI co-wrote a survey of LLMs for scientific research, including peer review (Luo et al., 2025), and the group's research-idea generator appears at EMNLP 2026 (Li et al., 2026a).
- *Adversarial evaluation.* The PI co-advised UT Dallas's team, one of the five university red teams in the Amazon Nova AI Challenge (Sahai et al., 2025), which built the COMET red-teaming framework (Xu et al., 2025), and holds an Open Philanthropy grant on control evaluations of AI debate.

This work has appeared at NeurIPS, ICLR, ACL, EMNLP, and TMLR.

5 Responsibility

Resources. UT Dallas provides the PI's laboratory space, a GPU server (six A100 and four RTX A6000 GPUs), and the shared Juno cluster (67 GPUs; see Facilities and Other Resources). With the requested cloud and model-access costs, these suffice; no equipment purchase is requested.

Ability to comply. The UT Dallas Office of Research and Innovation administers the PI's federal awards and will administer this award under 2 CFR 200 and ONR terms; the PI's commitments, listed in Current and Pending Support, leave the proposed effort available.

Performance history. The PI has managed an NSF CAREER award (2024–2029) and industry and foundation awards, filing reports on time; funds on current awards are being spent on schedule.

Integrity and business ethics. The PI and UT Dallas follow federal research-integrity policies (DoDI 3210.7); the PI has no record of misconduct.

Qualifications and eligibility. The PI is a U.S. permanent resident, has held a full-time tenure-track appointment at UT Dallas since 2022, the PI's first, and received the Ph.D. from Cornell University in 2021, on or after January 1, 2019, as Section B of the NOFO requires.

Organization and controls. UT Dallas maintains federally reviewed accounting, property-control, research-security, quality-assurance, and safety systems and is registered in SAM [UEI and CAGE code]; the research is computational.

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Principal Investigator's Curriculum Vitae

[Insert the PI's full CV here: education, appointments, awards and honors, grants, service, students, and all publications, including journal articles, conference papers, abstracts, book chapters, and patents, as NOFO Section D.1.c requires. Not counted toward the page limit.]

Letter(s) of Support

[Insert at least one letter from UT Dallas officials stating the university's long-term commitment to the PI and the proposed research. Not counted toward the page limit.]

Data Management Plan

Types of data, software, and other materials. The project will produce (1) three testbeds of questions with known answers: math and code problems with planted shared mistakes (A), multi-document questions composed from event structures, including synthetic maritime scenarios (B), and scientific-claim and peer-review questions with planted errors (C); (2) logs of agent answers, ratings, debates, tool calls and their outputs, judge rulings, and audit outcomes; (3) software: the agentic debate harness, the shared-error-factor rater model, the impartial selection mechanism, the audit scheme, red-team agents, reinforcement-learning code, and the simulation code behind the theoretical results; and (4) fine-tuned weights of open-weight models, where the original licenses permit release.

How the data will be acquired. Testbed A draws on public benchmarks (GSM8K and MATH, released under the MIT license) and public programming problems. Testbed B is generated by our MEQA pipeline from event structures; its maritime scenarios are fully synthetic and unclassified. Testbed C uses public papers and the public reviews and rebuttals posted on OpenReview, under the platform's terms, together with published reproducibility reports. Logs are produced by running agents built on open-weight models on UT Dallas computers and on commercial models through their interfaces. No human-subjects data are collected; in experiments, human audits are simulated by revealing known answers.

Time and location of acquisition. Not scientifically pertinent: all data are generated or collected computationally at UT Dallas during the 36-month project.

Processing. Questions are normalized to a common schema, deduplicated, and split into development, evaluation, and held-out sets by topic and by model family. Planted cases are generated by versioned scripts that record each edit and its effect on the answer. Logs are stored as produced and then summarized for analysis; every number in a publication is regenerated by a released script from the stored logs.

File formats and naming conventions. Questions and logs are stored as JSON Lines and Parquet; code is Python, released under an open-source license (MIT or Apache 2.0); model weights use the safetensors format. File names give the testbed, split, and version (for example, `tbB_eval_v1.2.jsonl`), with semantic versioning and a change log for each release.

Quality assurance and quality control. The known answer of every generated question is confirmed by construction and by an independent script; two students review a random sample of each release, and disagreements are resolved before release. Each log record carries the model version, prompt version, random seed, and a hash of tool outputs, so that any debate can be re-run. The harness is frozen at each milestone, and automated runs confirm that frozen versions reproduce stored results.

Origin of existing data. GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), the MEQA benchmark and pipeline (Li et al., 2024b), public OpenReview reviews, published reproducibility reports, and PeerEval logs produced by the PI's group. Each release records the source and license of every component.

Standards for data and metadata. Datasets will carry Croissant machine-learning metadata and a datasheet describing their purpose, composition, collection, and known limitations; code repositories will include environment files that pin software versions.

Preservation and access. Code will be public on GitHub, with each milestone release archived on Zenodo under a DOI. Testbeds and logs will be released on Hugging Face and archived in the Texas Data Repository [confirm UT Dallas repository] for at least five years after the project ends. Final reports will be submitted as required by DoD Instruction 3200.12.

Balance of value and cost. We will keep all data behind published results. Raw logs of exploratory runs, which can be large, will be kept for one year and then reduced to summaries, since the released code can regenerate them.

Security. All data are public or synthetic, unclassified, and contain no controlled unclassified information or personally identifiable information. If any data were later found to raise national security concerns, they would not be released, in accordance with DoD Instruction 5230.09.